NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Trauma

1. hip and femoral fractures
2. pelvis and acetabulum
3. tibial plateau and pilon

Hip and femoral fractures

1. Applied anatomy of the proximal femur

- **Neck-shaft angle** about 125-135 degrees; **anteversion** about 10-15 degrees in adults.
- **Blood supply of the femoral head:** mainly the **lateral epiphyseal (retinacular) branches of the medial femoral circumflex artery (MFCA)**, running up the posterosuperior neck. Lesser contributions: inferior gluteal, LFCA (anterior neck), and the artery of the ligamentum teres (small in adults; important in children). **Intracapsular fractures** disrupt the retinacular vessels: **AVN and non-union**.
- **Calcar femorale:** dense posteromedial cortical bone (load-bearing); **Ward's triangle** is the weak area between trabecular groups (Singh index grades osteoporosis from trabecular patterns).
- The **intertrochanteric line and capsule:** the capsule attaches anteriorly to the intertrochanteric line, but posteriorly only halfway down the neck.

2. Femoral neck fractures

Classifications

Garden	Description	Practical grouping
I	Incomplete / **valgus impacted**	Undisplaced
II	Complete, undisplaced	Undisplaced
III	Complete, **partially displaced** (trabeculae misaligned)	Displaced
IV	Complete, **fully displaced** (trabeculae of head realign with the acetabulum)	Displaced

- **Pauwels** (angle of the fracture line from horizontal; for young, high-energy fractures): **I under 30 degrees**, **II 30-50 degrees**, **III over 50 degrees (vertical shear, highest non-union and fixation failure)**.
- Location: subcapital, transcervical, basicervical (behaves like an extracapsular fracture).

Elderly (low-energy)

- **Undisplaced (Garden I-II): cannulated screws** (3 parallel screws in an inverted triangle, the inferior screw on the calcar, posterior screw against the posterior cortex; no screw entry below the lesser trochanter level: subtrochanteric fracture risk) or a sliding hip screw. Some evidence favours arthroplasty in very elderly frail patients with posterior tilt over about 20 degrees (high failure).
- **Displaced (Garden III-IV): arthroplasty**:
  - **Total hip arthroplasty** for active, independent, cognitively intact patients (better function; **HEALTH trial** showed small functional gains with more dislocations; NICE recommends THA for those who walk independently outdoors with a stick or less, are not cognitively impaired and are medically fit).
  - **Hemiarthroplasty** (bipolar or unipolar) for frailer, less active patients. **Cemented stems** are recommended (less pain, fewer periprosthetic fractures, better function: **WHiTE 5**), with care for **bone cement implantation syndrome**.
- **Orthogeriatric care:** surgery **within 36-48 hours** (NICE: on the day of or day after admission), early mobilisation, delirium prevention, nutrition, bone health (**secondary fracture prevention**: bisphosphonate, vitamin D), falls assessment. **One-year mortality about 25-30%.**

Young (high-energy)

- An **emergency**: **urgent anatomical reduction** (closed or open, Watson-Jones or Smith-Petersen approach) and **stable fixation** (cannulated screws, sliding hip screw ± derotation screw, or a fixed-

angle device such as a **dynamic hip screw with anti-rotation screw** or **proximal femoral locking/medial buttress plate** for vertical Pauwels III fractures).

- Complications: **AVN (about 20-30%)**, **non-union (about 10-30%)** (valgus intertrochanteric osteotomy (**Pauwels osteotomy**) converts shear into compression for non-union in young patients).
- **Ipsilateral femoral neck and shaft fracture:** the neck fracture (vertical, basicervical, undisplaced) is **missed in about 20-30%**: get a dedicated CT/fine-cut through the neck. Prioritise the **neck** (anatomical reduction).

Stress fractures of the femoral neck (Fullerton/Devas)

- **Compression (inferomedial)**: protected weight bearing; fix if the line exceeds about 50% of the neck or does not heal.
- **Tension (superolateral)**: high risk of displacement: **fix with screws**. Female athlete triad / RED-S.

3. Intertrochanteric (pertrochanteric) fractures

- Extracapsular; good blood supply, **unions reliably**; problem is fixation failure (**cut-out**).
- **Classification (AO/OTA 31A):**
  - **A1:** simple pertrochanteric (two-part; stable; intact lateral wall).
  - **A2:** multifragmentary pertrochanteric (posteromedial comminution, lateral wall incompetent in A2.2-2.3 per 2018).
  - **A3: intertrochanteric/reverse oblique** (unstable).
  - Evans (stable vs unstable; reverse obliquity is unstable), Boyd and Griffin, Jensen.
- **Unstable features: posteromedial comminution** (lesser trochanter), **reverse obliquity**, **subtrochanteric extension**, **lateral wall fracture** (thickness under about 20.5 mm predicts failure; Hsu), and inability to reduce.
- **Implants:**

  - **Sliding (dynamic) hip screw (DHS)** for **stable** fractures (A1): controlled collapse along the barrel.
  - **Cephalomedullary nail** (PFN, PFNA, Gamma, TFN, InterTAN) for **unstable** fractures, **reverse oblique** and subtrochanteric extension, deficient lateral wall. **DHS is contraindicated in reverse obliquity** (the fracture slides medially along the lag screw axis).
- **Tip-apex distance (TAD, Baumgaertner):** sum of the distances from the lag screw tip to the head apex on AP and lateral views, corrected for magnification. **Keep under 25 mm** (cut-out risk rises sharply above 25 mm). Position **centre-centre or inferior-centre** on AP, centre on lateral.
- **Complications: cut-out** (commonest mechanical failure; poor TAD, superior/anterior screw placement, unstable pattern), **Z-effect** (two-screw nails), **varus collapse**, periprosthetic fracture at the tip of a short nail, non-union (rare), malrotation.

4. Subtrochanteric fractures

- Within **5 cm distal to the lesser trochanter**. **High stress concentration** (medial compression, lateral tension).
- Deforming forces on the proximal fragment: **flexion (iliopsoas), abduction (gluteus medius/minimus) and external rotation (short external rotators)**; the shaft is pulled into **adduction (adductors)** and shortening.
- **Russell-Taylor** (piriformis fossa and lesser trochanter involvement); Seinsheimer.
- **Treatment: cephalomedullary (reconstruction) nail**, often after **open or clamp-assisted reduction** (reduction is the key; a malreduced varus fracture fails). Plates (95-degree blade plate, locking plates) in selected cases.
- **Atypical femoral fractures (AFF) with long-term bisphosphonates (ASBMR criteria):** subtrochanteric or diaphyseal, **minimal or no**

trauma, **transverse or short oblique** with a **medial spike** when complete, **non-comminuted**, **lateral cortical "beaking"/thickening**, prodromal thigh pain, often **bilateral** (image the other femur). **Stop the bisphosphonate**, calcium/vitamin D, consider **teriparatide**; **prophylactic intramedullary nailing** for incomplete fractures with pain or a visible fracture line.

5. Femoral shaft fractures

- High-energy injuries; blood loss up to about 1-1.5 litres; **fat embolism** risk; associated injuries (ipsilateral neck fracture, knee ligament injuries, hip dislocation).
- **Winquist-Hansen** comminution grades 0-IV.
- **Treatment of choice: reamed, statically locked antegrade intramedullary nail** within 24 hours in a stable patient. **Early fixation reduces pulmonary complications** (ARDS, fat embolism) and allows mobilisation.
- **Antegrade nail entry: piriformis fossa** (straight nails) or **trochanteric** entry (curved-proximal nails; avoids iatrogenic neck damage; small abductor damage). **Retrograde nail** (through the intercondylar notch): obesity, bilateral fractures, **ipsilateral femoral neck fracture** (with separate neck fixation), floating knee, pregnancy, distal fractures, ipsilateral acetabular fracture.
- **Reaming** increases union rates (**SPRINT for tibia** showed no harm; for femur, reamed nails union better); theoretical fat embolism risk.
- **Damage control orthopaedics (DCO): external fixation first**, nail later (within about 2 weeks; ideally day 4-14 when the inflammatory "second hit" settles), in **borderline or unstable** polytrauma patients (see polytrauma notes).
- **Complications:**
  - **Malrotation** (commonest malalignment after nailing; compare with the other side; over about 15 degrees is significant).

- **Limb-length discrepancy**, **non-union** (about 1-5%; exchange nailing for aseptic hypertrophic non-union), malunion (valgus with proximal fractures, apex-anterior with distal entry).
- **Fat embolism syndrome:** 24-72 hours, **Gurd and Wilson criteria** (major: respiratory insufficiency, cerebral signs, **petechial rash** (axilla, conjunctiva)); supportive care.
- Pudendal nerve palsy (traction table perineal post), heterotopic ossification at the entry, knee pain (retrograde).

6. Distal femur fractures

- **AO/OTA 33A (extra-articular), B (partial articular; B3 = coronal "Hoffa" fracture), C (complete articular).**
- **Hoffa fracture:** coronal plane fracture of a femoral condyle (**lateral more common**); seen only on lateral views/CT; fix with **anteroposterior lag screws** (perpendicular to the fracture).
- **Treatment: lateral locking plate** (MIPO) or **retrograde nail** (extra-articular and simple articular); **dual plating (lateral + medial)** for comminuted, bone loss or osteoporotic fractures and in nonunions; **distal femoral replacement** in elderly osteoporotic patients with very distal or periprosthetic fractures.
- **Deforming forces:** the **gastrocnemius** pulls the distal fragment into **extension (apex posterior)**; quadriceps and hamstrings shorten.
- **Periprosthetic distal femur fracture above a TKA:** Lewis-Rorabeck (I undisplaced, II displaced with a stable prosthesis, III loose prosthesis) or Su classification; fix (plate, retrograde nail if the box allows) or revise/replace if loose.

7. Patella (often with knee trauma)

- Undisplaced with intact extensor mechanism (can straight-leg raise): cylinder cast/brace.

- **Displaced (step-off over 2 mm or gap over 3 mm) or extensor lag: tension-band wiring** (K-wires + figure-of-eight; converts tension to compression) or cannulated screws with tension band; partial patellectomy for comminuted inferior pole.
- **Bipartite patella:** superolateral (Saupe type III), bilateral, smooth rounded margins: not a fracture.

Exam pearl: **Garden III-IV in the elderly = arthroplasty (cemented; THA if active and cognitively intact). Young displaced neck fracture = emergency anatomical reduction. Pauwels III = vertical shear, highest failure.**

Exam pearl: **TAD under 25 mm prevents cut-out. DHS for stable; nail for unstable, reverse oblique and subtrochanteric. Atypical femoral fracture: lateral cortical beaking, transverse, bilateral: stop bisphosphonates, consider teriparatide, nail.**

Exam pearl: **Ipsilateral neck + shaft fracture: the neck is often missed; prioritise the neck.**

Pelvis and acetabulum

1. Applied anatomy

- The pelvic ring is stable because of its **posterior ligamentous complex**: the **posterior sacroiliac ligaments (strongest ligaments in the body)**, interosseous SI ligaments, sacrospinous (resists external rotation) and sacrotuberous (resists vertical shear and flexion) ligaments, and the iliolumbar ligaments. **Posterior stability matters more than anterior.**
- Anterior: pubic symphysis (fibrocartilage); normal symphyseal width under about 5 mm, SI joint under about 4 mm.
- **Bleeding:** about 80-90% is **venous** (presacral and prevesical plexus) or from cancellous bone; about 10-15% is **arterial**, most commonly the **superior gluteal artery** (posterior, at the greater sciatic notch), then internal pudendal and obturator. The **corona mortis** (anastomosis between external iliac/inferior epigastric and obturator vessels, behind the superior pubic ramus, about 4-6 cm from the symphysis) is at risk in the Stoppa and ilioinguinal approaches.
- **Nerves:** the **L5 root** crosses the sacral ala (at risk with iliosacral screws placed too anteriorly); sacral roots in zones II/III sacral fractures; the lumbosacral trunk.

2. Pelvic ring injuries

Radiographs and CT

- AP pelvis, **inlet view** (beam angled caudally about 25-45 degrees: shows **AP displacement and rotation**) and **outlet view** (beam angled cephalad about 40-45 degrees: shows **vertical displacement** and sacral foramina). **CT** is standard.
- **L5 transverse process avulsion** suggests vertical instability. **Ischial spine avulsion** suggests sacrospinous disruption.

Young-Burgess (mechanism-based; predicts resuscitation needs)

Type	Mechanism	Injury pattern	Notes
APC I	Anteroposterior compression	Symphysis widening **under 2.5 cm**, SI ligaments intact	Stable; non-operative
APC II	AP compression	Symphysis **over 2.5 cm**, **anterior SI ligaments torn**, posterior SI intact ("open book")	Rotationally unstable, vertically stable
APC III	AP compression	Anterior and **posterior SI ligaments torn** (complete SI disruption)	Rotationally and vertically unstable; **highest blood loss and transfusion needs**
LC I	Lateral compression	Sacral buckle/impaction + oblique (horizontal) rami fractures	Stable
LC II	Lateral compression	Plus **iliac wing (crescent) fracture** or posterior SI disruption	Rotationally unstable
LC III	Lateral compression	**"Windswept pelvis"**: LC on one side, APC (open book) on the other	Unstable
VS	Vertical shear	Vertical displacement of a hemipelvis through SI joint, sacrum or ilium	Completely unstable; ipsilateral L5 transverse process avulsion
CM	Combined mechanism	Mixed patterns	

- **Lateral compression is the commonest pattern** (often elderly, side impact); **APC is associated with more bleeding**; **LC with head and**

chest injuries.

Tile (stability-based)

- **A:** stable (avulsions, iliac wing, transverse sacrococcygeal).
- **B:** rotationally unstable, vertically stable (B1 open book, B2 ipsilateral LC, B3 contralateral/bucket-handle).
- **C:** rotationally **and vertically unstable** (C1 unilateral, C2 bilateral, C3 with acetabular fracture).

Sacral fractures (Denis zones)

Zone	Location	Neurological injury
I	Lateral to the foramina (ala)	About 6%; **L5 root**
II	Through the foramina	About 28%; sciatica (unilateral roots)
III	Medial to the foramina (central canal)	About 57%; **bowel, bladder and sexual dysfunction**; includes **U/H-type (spinopelvic dissociation)** fractures

Emergency management of the bleeding pelvis

1. **ATLS**, two large-bore lines, **massive transfusion protocol** (balanced 1:1:1), **tranexamic acid within 3 hours** (CRASH-2).
2. **Pelvic binder at the level of the greater trochanters** (not the iliac crests) to reduce volume in APC injuries; internally rotate and tie the legs. Avoid over-compressing LC injuries.
3. **FAST** to look for intraperitoneal bleeding.
4. **Haemodynamically unstable despite resuscitation: preperitoneal pelvic packing** (with external fixation) and/or **angioembolisation** (arterial blush on CT; embolisation is the choice for arterial bleeding); **REBOA** in selected centres. Laparotomy if FAST positive.

5. **Anterior external fixation** (iliac crest or **supra-acetabular pins**; supra-acetabular gives better control, risk to the **lateral femoral cutaneous nerve**) or a **C-clamp** for posterior instability (risk: penetration through the greater sciatic notch).

Definitive fixation

- **Symphysis: Pfannenstiel** incision, plating of the superior surface (open book APC II/III).
- **Posterior ring: iliosacral screws** (percutaneous; S1 and S2 corridors; need good inlet/outlet/lateral imaging; danger to **L5 root** anteriorly and **S1 root** through the foramen; **sacral dysmorphism** narrows the S1 corridor), anterior SI plating, posterior tension-band plating, **lumbopelvic fixation** for spinopelvic dissociation.
- **Fragility fractures of the pelvis (FFP, Rommens and Hofmann):** elderly, low-energy LC-type injuries with sacral involvement; mobilise; fix (percutaneous screws) if pain prevents mobilisation or the fracture progresses.

Associated injuries and complications

- **Urethral injury** (males, mostly **posterior (membranous-bulbar) urethra**, APC and straddle injuries): **blood at the meatus, high-riding prostate, perineal haematoma**: **retrograde urethrogram before a catheter**; suprapubic catheter.
- **Bladder rupture** (extraperitoneal commonest; repair intraperitoneal rupture surgically).
- **Open pelvic fractures** (perineal or rectal wounds): mortality up to about 50%; **diverting colostomy** for rectal/perineal wounds.
- **Morel-Lavallee lesion:** closed degloving over the greater trochanter (shearing off the fascia); infected easily; drain/debride (percutaneous or open) before or with fixation.
- **DVT/PE** (high risk; chemoprophylaxis), **neurological deficit** (L5, S1), sexual dysfunction, chronic SI pain, malunion with limb-length discrepancy and sitting imbalance.

3. Acetabular fractures

Columns and radiology

- **Anterior column:** iliac crest (anterior part) to the pubic symphysis, including the anterior wall. **Posterior column:** greater sciatic notch to the ischial tuberosity, including the posterior wall.
- **Judet views** (45-degree obliques): **obturator oblique** (shows the **anterior column** and **posterior wall**) and **iliac oblique** (shows the **posterior column** and **anterior wall**). Six radiographic lines on AP: **iliopectineal line (anterior column)**, **ilioischial line (posterior column)**, teardrop, roof, anterior rim, posterior rim.
- **Spur sign** (obturator oblique): the intact iliac bone above the joint projecting, seen in **both-column fractures** (no articular segment attached to the axial skeleton).
- **Roof arc angles (Matta):** medial, anterior and posterior roof arcs **over 45 degrees** mean the fracture spares the weight-bearing dome (non-operative may be possible).

Letournel classification

Elementary (5)	Associated (5)
Posterior wall (**commonest overall**)	**Both column** (commonest associated; no articular fragment attached to the ilium; spur sign)
Posterior column	T-type
Anterior wall	Transverse + posterior wall (common)
Anterior column	Posterior column + posterior wall
Transverse (only elementary type through both columns)	Anterior column/wall + posterior hemitransverse

- **Elderly:** anterior column / anterior wall patterns with **quadrilateral plate** medialisation and **superomedial dome impaction ("gull**

sign") are typical (low-energy falls).

Management

- **Hip dislocation with acetabular fracture: urgent reduction** (within about 6 hours to reduce AVN), check the sciatic nerve, skeletal traction if unstable.
- **Non-operative:** undisplaced (under about 2 mm), roof arcs over 45 degrees with a congruent joint out of traction, secondary congruence of both-column fractures, small posterior wall fragments with a stable hip (**stress EUA**), elderly patients unfit for surgery.
- **Operative indications: articular displacement over 2 mm** in the dome, **unstable hip** (posterior wall over about 40-50% of the wall is usually unstable; under 20% usually stable), **incarcerated intra-articular fragments**, marginal impaction. Fix within about 2 weeks (the window closes by about 3 weeks).
- **Approaches:**

Approach	Access	Structures at risk
Kocher-Langenbeck (posterior; patient prone or lateral)	Posterior wall and column, transverse, T	**Sciatic nerve** (keep hip extended, knee flexed), **superior gluteal vessels/nerve** at the notch, **medial femoral circumflex artery** (preserve quadratus femoris and obturator externus)
Ilioinguinal (Letournel; 3 windows)	Anterior column/wall, both column	**Femoral nerve and vessels**, **lateral femoral cutaneous nerve**, **corona mortis**, spermatic cord, inguinal hernia
Modified Stoppa (anterior intrapelvic)	Quadrilateral plate, anterior column, both column	**Corona mortis**, obturator nerve and vessels

Approach	Access	Structures at risk
Extended iliofemoral	Both columns (complex late cases)	Highest **heterotopic ossification** risk

- **Elderly:** fix-and-replace (acute THA with acetabular fixation) for severe dome impaction, femoral head damage or pre-existing arthritis.

Complications

- **Post-traumatic arthritis** (commonest late complication; related to the accuracy of reduction: anatomical reduction under 1 mm gives the best results).
- **Heterotopic ossification** (posterior and extensile approaches; prophylaxis with **indomethacin** or **single-dose radiotherapy**; Brooker classification).
- **AVN of the femoral head** (with posterior dislocation), **sciatic nerve palsy** (peroneal division most affected; about 10-15%, partly iatrogenic), **DVT/PE**, infection, malreduction.

4. Hip dislocation (often with acetabular injury)

- **Posterior (about 90%):** dashboard injury, hip **flexed, adducted and internally rotated**; associated posterior wall fracture, **femoral head fracture**, **sciatic nerve injury** (peroneal division), and **ipsilateral knee injury (PCL, patella)**.
- **Anterior:** flexed, abducted and externally rotated; femoral nerve and vessels at risk.
- **Reduce urgently (within 6 hours)**; then CT for intra-articular fragments and head/wall fractures. **AVN** risk rises with delayed reduction (about 5-10% overall).
- **Thompson-Epstein** (posterior dislocation types I-V) and **Pipkin** (femoral head fractures with posterior dislocation):

Pipkin	Description	Treatment
I	Head fracture **below (caudal to) the fovea** (non-weight-bearing)	Non-operative if reduced, or excise small fragments
II	Head fracture **above (cephalad to) the fovea** (weight-bearing)	ORIF (often anterior Smith-Petersen or surgical hip dislocation)
III	I or II + **femoral neck fracture**	ORIF of neck (young) or arthroplasty (older); worst prognosis (AVN)
IV	I or II + **acetabular fracture**	Fix the acetabulum and head

Exam pearl: **APC III = most bleeding; LC = commonest; superior gluteal artery = commonest arterial bleeder; binder at the greater trochanters.** Retrograde urethrogram before catheterising a male with blood at the meatus.

Exam pearl: **Obturator oblique shows anterior column + posterior wall; iliac oblique shows posterior column + anterior wall. Spur sign = both-column fracture.** Posterior wall = commonest acetabular fracture; Kocher-Langenbeck endangers the sciatic nerve and MFCA.

Tibial plateau and pilon

1. Tibial plateau fractures

Anatomy and mechanism

- The **lateral plateau** is **convex** and higher; the **medial plateau** is **concave**, larger and stronger (more load).
- Mechanism: **valgus load with axial force** (lateral plateau, commonest), varus (medial plateau, higher energy, often **fracture-dislocation**).
- **Lateral plateau fractures are commonest (about 55-70%).**
- Associated injuries: **lateral meniscus tears** (commonest with lateral plateau; often peripheral and incarcerated), **medial meniscus**, **MCL and ACL injuries**, **peroneal nerve** (lateral fracture-dislocation), **popliteal artery** (medial plateau and Schatzker IV/VI, bicondylar fractures; **ankle-brachial index under 0.9 = arterial study**), **compartment syndrome** (especially Schatzker VI).

Schatzker classification

Type	Description	Typical patient / mechanism	Treatment principle
I	Lateral **split** (wedge)	Young, dense bone	Percutaneous screws or buttress plate; check for meniscal entrapment
II	Lateral **split-depression**	**Commonest**	Elevate the depressed fragment, **subchondral bone graft/substitute**, raft screws, lateral buttress plate

Type	Description	Typical patient / mechanism	Treatment principle
III	Lateral **pure depression**	**Osteoporotic** elderly	Elevation through a cortical window, graft, raft screws (IIIa lateral, IIIb central)
IV	**Medial plateau**	High energy; fracture-dislocation (Moore); **popliteal artery and peroneal nerve at risk**	Posteromedial approach, buttress plate; poor prognosis
V	**Bicondylar** with metaphysis intact	High energy	Dual plating / lateral locking ± medial plate
VI	Bicondylar with **metaphyseal-diaphyseal dissociation**	Highest energy; **compartment syndrome**	**Staged**: spanning external fixator, then ORIF (dual incisions) when soft tissues allow; or hybrid/circular frame

- **Three-column concept (Luo; CT-based):** lateral, medial and **posterior** columns (the posterior column, especially **posteromedial fragments**, is missed by Schatzker; needs a **posteromedial (Lobenhoffer) or posterior approach** and buttress plating).
- **Moore classification** (fracture-dislocations): split, entire condyle, rim avulsion, rim compression, four-part.

Investigation

- AP, lateral, oblique; **CT** for every operative plateau; **MRI** shows meniscal/ligament injuries (commonly present).
- **Assess soft tissues** (fracture blisters, swelling: **"wrinkle sign"** shows readiness for surgery, usually 7-14 days).

Management

- **Non-operative:** undisplaced or minimally displaced, stable to varus/valgus stress in full extension (under about 10 degrees), elderly low-demand.
- **Operative indications: articular step-off over 2-3 mm** (debated: stability and alignment matter more), **condylar widening over 5 mm**, **instability (over 10 degrees varus/valgus in extension)**, **medial plateau fractures** (inherently unstable), open fractures, compartment syndrome, vascular injury.
- **Principles:** restore the **mechanical axis and joint stability** first (more important than perfect articular congruity), elevate the joint surface, **support it with subchondral rafting screws and bone graft/substitute** (calcium phosphate cement resists subsidence best), **buttress** the metaphysis, repair menisci.
- **Approaches:** anterolateral (lateral plateau; submeniscal arthrotomy to see the joint); posteromedial (Lobenhoffer, between the medial gastrocnemius and pes anserinus/semimembranosus); posterior (Burks-Schaffer, Carlson); avoid a single midline incision with extensive stripping for bicondylar fractures (**dual incisions with a skin bridge over about 7 cm** historically; soft tissue is the priority).
- **Staged protocol for high-energy fractures: spanning external fixation** (ligamentotaxis), CT in the frame, definitive fixation once the soft tissues settle.
- **Arthroscopy-assisted reduction (ARIF)** for lateral split/depression fractures.
- **Elderly with severe comminution/arthritis:** acute TKA in selected cases.

Complications

- **Infection and wound breakdown** (bicondylar, early surgery through poor soft tissue), **post-traumatic arthritis** (malalignment and instability more than step-off), **stiffness/arthrofibrosis**, loss of

reduction (subsidence), **compartment syndrome**, **peroneal nerve injury**, DVT, malunion.

2. Tibial shaft fractures (brief link)

- Commonest long-bone fracture; **highest rate of open fractures** and compartment syndrome.
- **Reamed intramedullary nail** for most displaced fractures (**SPRINT**: no difference between reamed and unreamed overall; possible benefit of reaming in closed fractures). **Suprapatellar (semi-extended) nailing** eases proximal and distal third fractures (less anterior knee pain debated).
- **Proximal third fractures** tend to **valgus and procurvatum (apex anterior)** with nailing: use **blocking (Poller) screws**, a more lateral/proximal start point, extended positioning, plates.
- **Distal third**: nail with blocking screws, or plate; fix the fibula if needed for alignment.
- Non-operative (Sarmiento functional bracing) for low-energy, stable fractures with acceptable alignment (under 5 degrees varus/valgus, under 10 degrees AP angulation, under 1 cm shortening, under 10 degrees rotation).
- **Anterior knee pain** is the commonest complication after tibial nailing.

3. Tibial pilon (plafond) fractures

Mechanism and features

- **Axial loading**: the talus is driven into the distal tibial articular surface (falls from height, road traffic accidents). Distinct from rotational ankle fractures.
- **Soft-tissue injury is the dominant problem** (thin anterior and medial soft tissue envelope).

- Fragments (Chaput's): typically **medial**, **anterolateral (Chaput, attached to AITFL)**, **posterolateral (Volkmann, attached to PITFL)**, with central die-punch impaction; associated fibula fracture in about 75%.
- **Associated injuries:** spine fractures, calcaneus fractures, contralateral injuries (falls from height).

Classification

Ruedi-Allgower	Description
I	Undisplaced cleavage fracture of the joint
II	Displaced articular fracture, minimal comminution
III	Comminuted and impacted articular surface

- **AO/OTA 43:** A extra-articular, B partial articular, C complete articular (C3 = comminuted articular + metaphyseal).
- **Tscherne** for closed soft-tissue injury (C0-C3).

Management: the staged protocol

1. **Stage 1 (urgent): spanning external fixator** (tibia to calcaneus/first metatarsal) to restore length, alignment and rotation by ligamentotaxis; ± **fibular fixation** (if the plafond team and soft tissue plan allow; malreduced fibula makes later tibial reduction impossible).
2. **CT in the frame**: plan approaches.
3. **Stage 2 (when soft tissues allow, usually 7-21 days; "wrinkle sign", blisters re-epithelialised): ORIF** of the articular surface with plates (anterolateral, anteromedial, posterolateral, medial approaches depending on fragments), bone graft for metaphyseal defects.
4. **Rüedi-Allgöwer principles:** (1) restore fibular length, (2) reconstruct the articular surface, (3) bone graft the metaphyseal defect, (4)

buttress the medial tibia.

5. **Alternatives:** definitive **circular (Ilizarov) or hybrid external fixation** with limited internal fixation when soft tissues are poor; **primary ankle arthrodesis** for unreconstructable fractures.
6. **Approaches: anterolateral** (between the EDL/peroneus tertius and the fibula; superficial peroneal nerve at risk), **anteromedial** (medial to tibialis anterior; thinnest soft tissue), **posterolateral** (between peroneals and FHL; sural nerve), **medial**. Skin bridges of at least 5-7 cm between incisions (debated).

Complications

- **Wound dehiscence and deep infection** (historically up to 30-40% with early ORIF; staged protocols reduced this to under 5-10%).
- **Post-traumatic ankle arthritis** (common; related to the initial cartilage damage more than reduction quality), malunion (varus), non-union (metaphyseal), stiffness, chronic pain, complex regional pain syndrome.

> Exam pearl: **Schatzker II (split-depression) is the commonest; IV (medial) = fracture-dislocation with popliteal artery and peroneal nerve risk; VI = metaphyseal-diaphyseal dissociation with compartment syndrome.** The **lateral meniscus** is the commonest associated soft-tissue injury.

> Exam pearl: **Pilon = axial load; soft tissue dictates timing; stage it: span, scan, plan, then fix (7-21 days).** Rüedi-Allgöwer's four steps start with **fibular length**.